Statistical Analysis of an Adversarial Bayesian Weak Supervision Method

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Abstract

Suppose one has labeling functions (LFs, e.g. crowdsourcing workers) that make predictions on unlabeled datapoints. A recent approach produces a polytope of plausible labelings by using the LF predictions and then picks the "center". While this approach performs well empirically, we are unaware of theoretical analysis for it. We propose a new approach that follows a similar strategy, also has strong empirical performance, and is amenable to theoretical analysis.

Formal Model

- N unlabeled datapoints $X = \{x_1, x_2, \dots, x_N\} \subset \mathcal{X}$
- K labels denoted $\mathcal{Z} = \{1, 2, \dots, K\}$
- W labeling functions (LF) that can each abstain:

$$h^{(1)}, h^{(2)}, \dots, h^{(W)} : X \to \mathcal{Z} \cup \{?\}$$

Goal: For each datapoint x_i and labels $1 \leq j \leq k$, infer the underlying conditional label probability

$$\eta_{ij} = \Pr(z_i = j \mid x_i).$$

A Geometric Approach: Hyper Label Model

For simplicity (and only in this section), assume that $\mathcal{Z} = \{\pm 1\}$. Construct a set of plausible labelings U for which at least $W/2$ LFs get accuracy $> 50\%$ on the set of points with label 1 and label 2 respectively. The Hyper Label Model (HyperLM) prediction is the minimizer to all elements of U, i.e.

$$\vec{g}^{hlm} = \frac{1}{|U|} \sum_{\vec{z} \in U} \vec{z} = \underset{\vec{g} \in [-1,1]^N}{\arg\min} \frac{1}{|U|} \sum_{\vec{z} \in U} \|\vec{z} - \vec{g}\|_2^2.$$

$\vec{g}^{hlm}$ can be thought of as the "center" of the convex hull of U.

Questions of Interest: How can we analyze such an approach given its strong empirical performance? What might another model using the geometric approach look like?

Our Approach: Bayesian Balsubramani-Freund (BBF)

Balsubramani-Freund selects the max entropy labeling from a polytope P of feasible labelings. Every labeling in P gives every LF $h^{(w)}$ accuracy in $[b_w - \epsilon_w, b_w + \epsilon_w]$ where ϵ_w is estimated via labeled data. What if we put a distribution on LF accuracies?

For brevity, define the following two sets of labelings:

- $\mathcal{L}_{\vec{\tau}}$, the set of labelings which have class distribution $\vec{\tau}$
- $\mathcal{L}_{b,\vec{z}}$, the set of predictions with accuracy b against labeling $\vec{z}$.

Here is BBF's (informal) generative process:

1. Draw class frequencies $\vec{\tau} \sim Dirichlet(\vec{\alpha})$, $\vec{\alpha} \in \mathbb{R}^K_{>0}$
2. Draw labels for all N datapoints $\vec{z} \sim Entropy(\mathcal{L}_{\vec{\tau}})$ where $Pr(\vec{z} \mid \vec{\tau}) \propto \exp(-\vec{z}^\top \log \vec{z})$
3. For each LF $w \in [W]$:
   a. Draw $h^{(w)}$'s accuracy $b_w \sim Beta(\vec{\rho}_w)$, $\vec{\rho}_w \in \mathbb{R}^2_{>0}$
   b. Draw $h^{(w)}$'s preds on all N points $\vec{y}_w \sim Uniform(\mathcal{L}_{b_w,\vec{z}})$

Unlike iBCC/EBCC/FABLE, we don't assume independence of an LF's predictions between different datapoints. Moreover, BBF attempts to model the class frequencies/LF accuracies on X rather than the *underlying* quantities.

Our Contributions

BBF's Posterior Label Distribution is Log-Concave

We take BBF's posterior mode as its prediction, which can be found by optimizing a concave program.

BBF is an Example of the Geometric Approach

One can define an implicit polytope centered around the posterior mode (BBF's prediction) upon inspection of BBF's log-posterior distribution.

Our Contributions (Cont.)

Functional Form of BBF Predictions

BBF draws its predictions from the same exponential family $\mathcal{G}$ as BF, i.e. it's the softmax of a weighted majority vote.

BBF's Consistency/Rates of Convergence

As the priors concentrate on the empirical LF accuracies, the BBF prediction converges to $\vec{g}^*$, the unique best approximator in $\mathcal{G}$ for the underlying conditional label probabilities η. We bound how fast BBF's prediction tends to $\vec{g}'$.

Select Experimental Results for BBF

BBF was compared against 12 baseline methods on 11 datasets from the WRENCH benchmark. The # of classes ranged from 2 to 10, LFs 4 to 164, and datapoints $1.8k$ to $82k$.

- BBF matches or outperforms FABLE, the best Bayesian approach on all but one dataset.
  - Unlike BBF, FABLE uses datapoint features. Also, FABLE models each LF using $3K^2$ parameters while BBF uses 1.
- BBF is competitive with HyperLM, the best method that doesn't use datapoint features on $10/12$ datasets.
- When the conditions for the consistency result are simulated on real datasets, BBF's prediction tends to $\vec{g}'$ in KL divergence.

Takeaways

- The theoretical analysis of BBF shows how a method of the geometric approach can be analyzed.
- BBF is an example of an adversarial method that is both unsupervised and has nice theoretical properties. To our knowledge, previous proposals had one or the other.
- There's more to be done with methods that don't use datapoint features – BBF outperforms two out of three methods considered that use datapoint features.